Developer Workstation - Configuring your Docker

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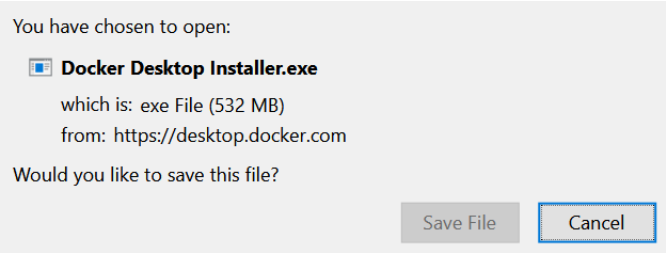
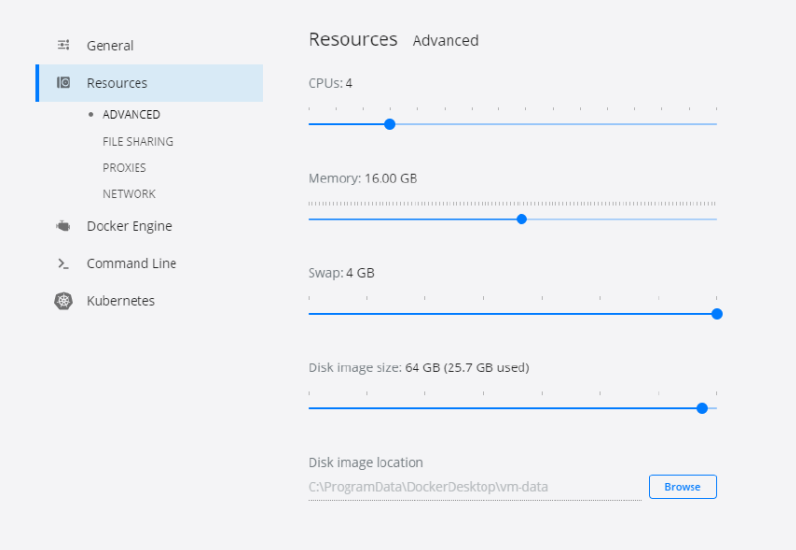
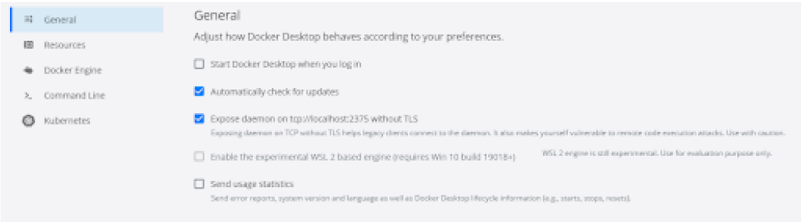
Scope of Page: (Currently in DRAFT Form and should be considered a fluid document)

To provide any given developer the necessary steps to stand up their Developer Workstation for this effort.

Steps	Instructions	Description / Comments
1	Download the latest version of Docker	⚠ As of this date, 31 May 2021 , Docker is within an Allstate PEAP Process, to receive permission to be used ON PREMISE. At this current stage, a downloading the software is possible, however the virtualization required for Docker; whether it be the "**Oracle VM**" or "**Microsoft-Hyper-V**" or another compatible software, is currently deemed a Security Risk. Without approval will not be able to move forward. These steps will be treated as fluid, until such approval is achieved ON PREMISE. As before, download the 64 bit version, of Docker; https://www.docker.com/get-started 
2	Double Click the Docker Desktop Icon & Make the following Configuration changes.	⚠ Docker needs at least 16GB of RAM to actually start up the services required, 18GB for some tasks. Go to Docker settings and change the allocatable memory.  **<u>Text Version of Configurations:</u>** ***Note***: These settings may also be found at the following relative path ***<u>(%UserProfile%\AppData\Roaming\Docker)</u>*** 1. **CPU's:** 4 2. **Memory**: 16.00 Gb's

		3. **Swap**: 4 Gb's 4. **Disk Image Size**: 64 Gb's 5. **Disk Image Location**: '*C:\insertSystemFolderPathHere*'
3	Configure the General Section of Docker	'Check' the option to expose daemon without the ***TLS***: 
4	Configure Docker Hub	Docker links are configured to be using EISU Docker/Maven repositories. They can be reconfigured if not working in an EISU environment to a location-specific or project-specific hub. To be able to download Docker images for applications, find all Dockerfile files and change from (actual links can be different): ARG GENESIS_BASE_IMAGES=dev5eisunexus01.eqxdev.exigengroup.com:5000/genesis-base:1.4 to ARG GENESIS_BASE_IMAGES=<dockerHostWithPort>/<pathToImages>/genesis-base:1.4
5	Configure YML Files	Also find *.yml files in ref-impl/docker and replace the existing (i.e. dev5eisunexus01) docker url for images with your docker url in the same manner. In Windows, you can do it faster with an editor like Notepad++ using Replace in files function (or you can try rpl for windows alternative from http://gnuwin32.sourceforge.net/packages/rpl.htm or choco install gnuwin -y) In Linux, either command must exist already or can use apt install rpl to install it. There is a replace.bat file in Docker folder which you can use to replace Docker links, that is: rpl.bat dev5eisunexus01.eqxdev.exigengroup.com:5000 <dockerHostWithPort>/<pathToImages>
6	Validate the Docker Installation was Successful	Open a Command Line, as Elevated Privileges, and execute the following Docker Command; "**docker run -d -p 80:80 docker/getting-started**" After it runs you can open http://localhost/ and check the documentation for success
7	Continued Validation - Log in to Docker	If you have not already done so, Open a Command Line, as Elevated Privileges. Execute the following Docker Command; "**docker login -u <username> -p <password> <dockerHostWithPort>/<pathToImages>**" Successful login must display: **Login Succeeded**
	Trouble Shooting	If login fails, it will not write credentials to Windows Credential Manager. To check if you have access rights, you can open this link from a browser and providing with your credentials it should show the tags list: https://dev5eisunexus01.eqxdev.exigengroup.com:5000/v2/cap-case-facade/tags/list
		Also you can check ~/.docker/config.json or %UserProfile%/.docker/config.json to get a list of servers where you are logged in. You can also Docker logout from them Make sure Docker login is successful. Check if Docker is working correctly: • Check Docker through the web (GET https://<dockerHostWithPort>/v2/) and it must return the 200 OK response. • Check docker for image list GET https://<dockerHostWithPort>/v2/_catalog) - there should be cap-* images. • Check docker for image tags GET https://<dockerHostWithPort>/v2/<pathToImages>/cap-case-facade/tags/list or GET • https://<dockerHostWithPort>/v2/cap-case-facade/tags/list - there should be latest • Check docker for manifest of specific image and version, i.e. https://<dockerHostWithPort>/v2/<pathToImages>/cap-case-facade • /manifests/11.5 or https://<dockerHostWithPort>/v2/cap-case-facade/manifests/11.5 - should just show json manifests.
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9	Intentionally left blank	Intentionally left blank, for future plausible steps

10	**Congrats!!!, you have successfully installed and configured Docker**

No labels